01

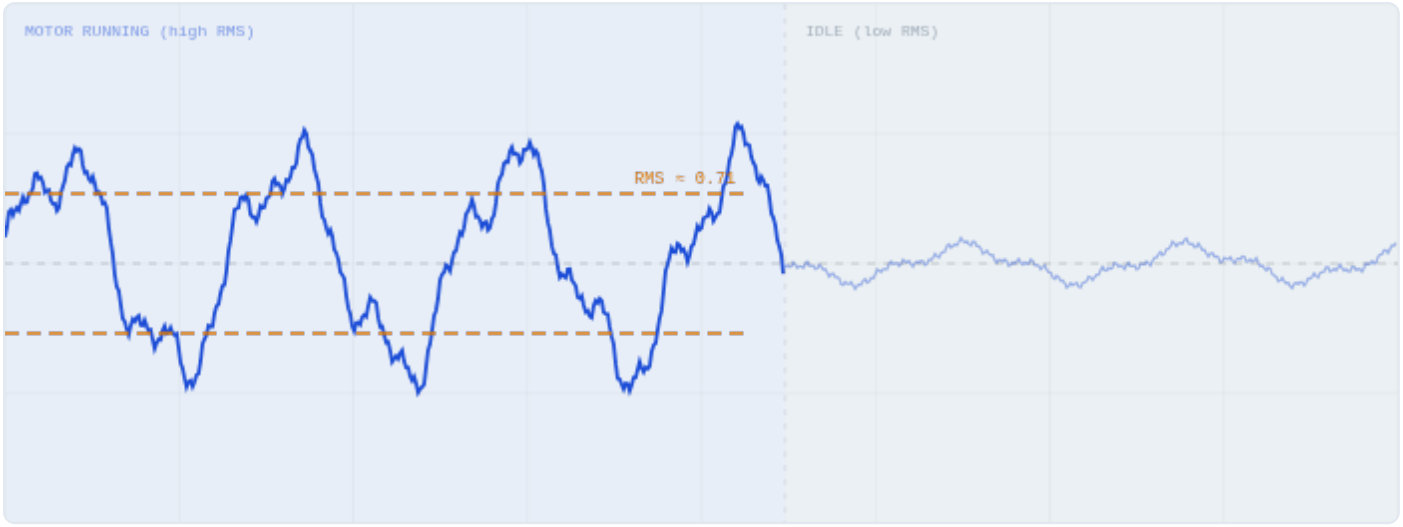
Root Mean Square (RMS)

Effective energy or "loudness" of the signal over a time window

LOAD INDICATOR

FORMULA $RMS = \sqrt{\frac{1}{N} \cdot \Sigma x_i^2}$

N = window length · x_i = i-th acceleration sample



TIME DOMAIN · AMPLITUDE VS TIME · MOTOR RUNNING ZONE VS IDLE ZONE

RMS measures the **effective power** of the vibration signal across a sample window. It acts as the first gating indicator — a near-zero RMS confirms the motor is **Off**, while a high stable RMS confirms activity. However, loud ambient noise also elevates RMS identically to a running motor, which is why the remaining five features are essential for correct classification.

STATE	TYPICAL RMS	INTERPRETATION
● **Healthy Motor**	0.20 – 0.80	Normal load, stable rotation
● **Mechanical Fault**	> 1.20	Overload or vibration surge detected
● **Crowd / Ambient**	0.40 – 1.50	Easily confused — requires ZCR gating
● **Dead Silent**	< 0.05	Motor off, only sensor thermal noise

02

Kurtosis

Statistical "peakiness" of the amplitude distribution — identifies impulsive events

IMPULSE DETECTOR

FORMULA $K = [\ 1/N \cdot \Sigma\ (x_i - \mu)^4\ ]\ /\ \sigma^4$ μ = mean · σ = std deviation



TIME DOMAIN · HEALTHY SIGNAL (K ≈ 3) VS MECHANICAL FAULT (K > 10) — IMPULSE SPIKES SHOWN IN RED

The **gold standard for mechanical looseness detection**. A healthy motor produces a near-Gaussian vibration distribution with Kurtosis of ~3. Repetitive mechanical impacts — a loose screw rattling, a cracked bearing — generate sharp, periodic spikes that drive Kurtosis above 10. The **variance of Kurtosis over time** is equally important: low variance at high K means structural fault; high variance means chaotic ambient noise.

STATE	KURTOSIS VALUE	VARIANCE OVER TIME	INTERPRETATION
● **Healthy Motor**	2.5 – 3.5	Low	Gaussian distribution, stable
● **Mechanical Fault**	> 10	Low (persistent)	Repetitive impulses — looseness
● **Crowd / Ambient**	3.0 – 6.0	**High**	Chaotic, non-structural spikes
● **Dead Silent**	~3.0	Minimal	Residual sensor thermal noise

03

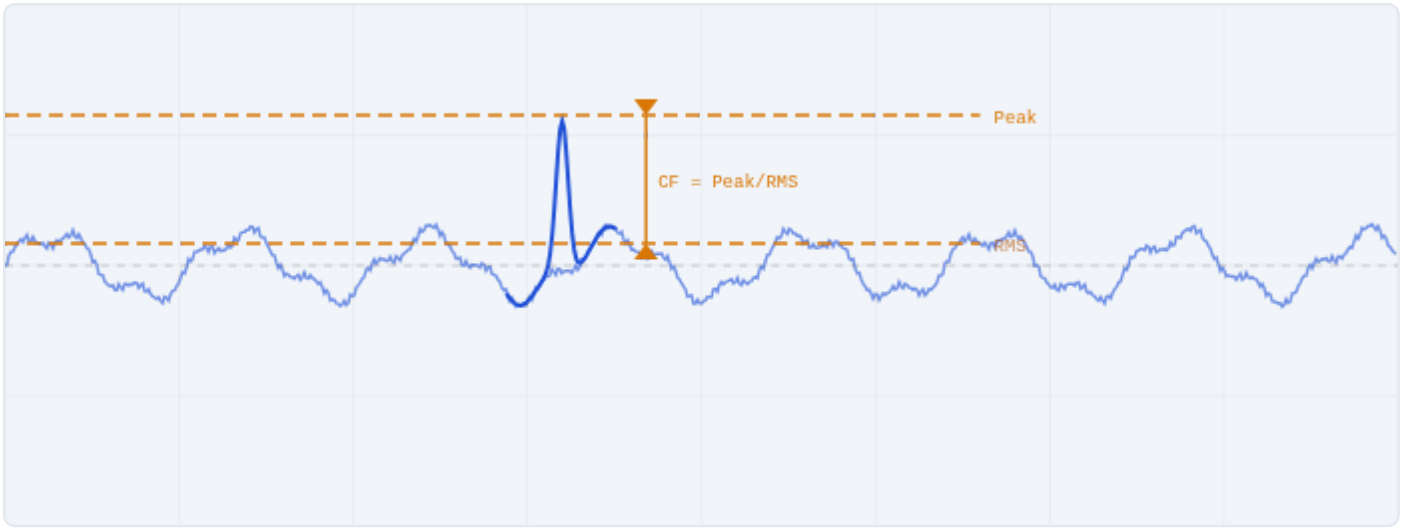
Crest Factor

Ratio of peak amplitude to RMS — sensitivity to sudden, non-repetitive shocks

SHOCK DETECTOR

FORMULA `CF = max |x| / RMS(x)`

Pure sine wave: CF = √2 ≈ 1.414 (theoretical minimum)



TIME DOMAIN · LOW RMS BASELINE WITH A HIGH-AMPLITUDE SHOCK PULSE — PEAK / RMS RATIO ANNOTATED

Identifies **sudden, non-repetitive shocks** that Kurtosis might smooth over in averaged windows. A quiet baseline with one large spike keeps RMS low but drives the Peak high, producing a very large CF. A healthy motor at constant load has a stable CF near 1.4. **Early bearing spalling** elevates CF significantly before RMS increases — making this the earliest mechanical fault indicator alongside Spectral Centroid.

STATE	CREST FACTOR	INTERPRETATION
● **Healthy Motor**	1.4 – 2.5	Steady sinusoidal load, RMS ≈ Peak/√2
● **Early Bearing Fault**	> 6.0	Spalling impulses, RMS still low
● **Severe Fault**	3.0 – 5.0	RMS rises, CF ratio compresses
● **Single Drop / Shock**	> 10	Very high peak, near-zero baseline RMS

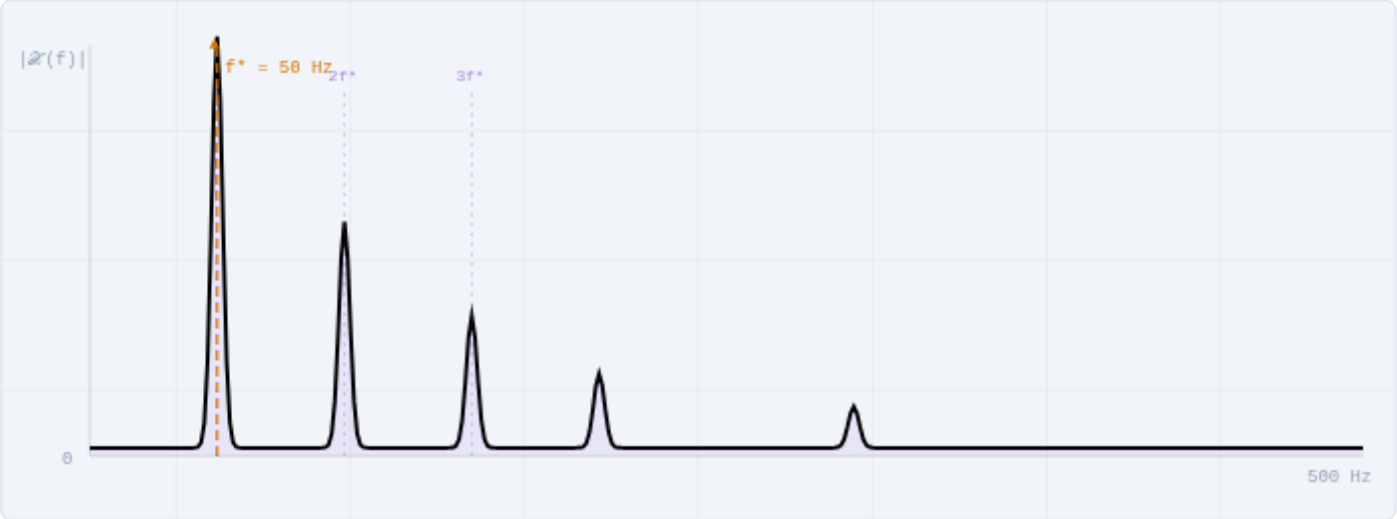
04

Peak Frequency

Dominant frequency via Fast Fourier Transform — the motor's operating "heartbeat"

CONTEXT SWITCHER

FORMULA $f^* = \text{argmax}_f |\mathcal{X}(f)| \quad \text{where} \quad \mathcal{X}(f) = \Sigma\, x_n \cdot e^{-j2\pi fn/N}$ — $\mathcal{X}(f)$ = FFT output



FREQUENCY DOMAIN · FFT MAGNITUDE SPECTRUM · FUNDAMENTAL PEAK F* AND HARMONIC FAMILY LABELED

The motor's **operating frequency fingerprint**. Each supply voltage produces a distinct fundamental frequency — at 6V the motor hums near 45 Hz; at 8V it shifts to ~65 Hz. By locating the dominant FFT peak, the classifier can **switch operating context** and compare subsequent features against the correct voltage-mode baseline. Ambient noise produces no narrow dominant peak — it spreads energy broadly across the spectrum.

STATE	PEAK FREQUENCY	SPECTRAL SHAPE	INTERPRETATION
● **Motor @ 6V**	40 – 55 Hz	Narrow, stable	Consistent harmonic family
● **Motor @ 8V**	60 – 75 Hz	Narrow, stable	Shifted harmonic family
● **Mechanical Fault**	f* ± sidebands	Peak + flanking bands	Fault frequency modulation
● **Ambient Noise**	Broad / undefined	Flat noise floor	No dominant narrow peak

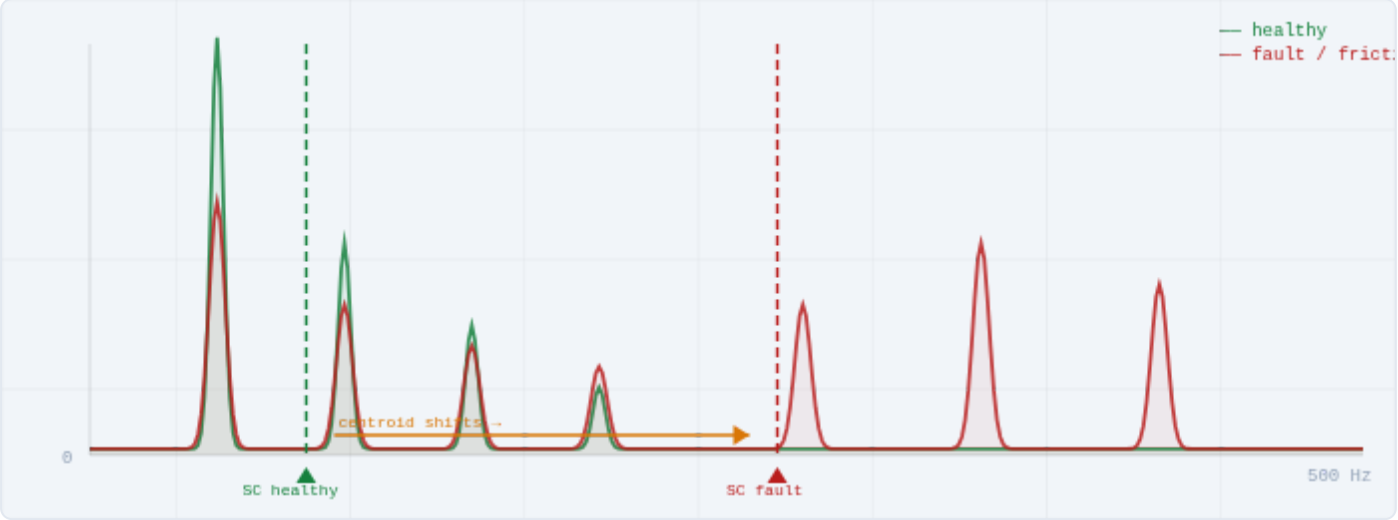
05

Spectral Centroid

Center of mass of the frequency spectrum — the perceived "brightness" of vibration

FRICTION DETECTOR

FORMULA $SC = \Sigma\ f_k \cdot |\mathscr{X}(f_k)| \ / \ \Sigma\ |\mathscr{X}(f_k)|$ Weighted average frequency by spectral magnitude



FREQUENCY DOMAIN · HEALTHY SPECTRUM (DARK CENTROID) VS FAULT SPECTRUM (BRIGHT, HIGH-FREQ CENTROID)

A bearing or shaft experiencing **friction generates a high-frequency hissing** sound long before the motor physically seizes. This shifts the spectral centroid upward — the spectrum becomes "brighter." A healthy motor has a **dark, low centroid** concentrated near the fundamental frequency. The Spectral Centroid is the **earliest available indicator** of bearing degradation, typically rising before Kurtosis crosses its fault threshold.

STATE	CENTROID POSITION	INTERPRETATION
● **Healthy Motor**	Low — near f*	Energy focused at fundamental + harmonics
● **Early Friction**	Moderately elevated	Broadband hiss beginning to emerge
● **Bearing Failure**	High — > 3× healthy	High-frequency noise now dominant
● **Speech / Crowd**	Very high (~1–4 kHz)	Wideband speech energy

06

Zero Crossing Rate (ZCR)

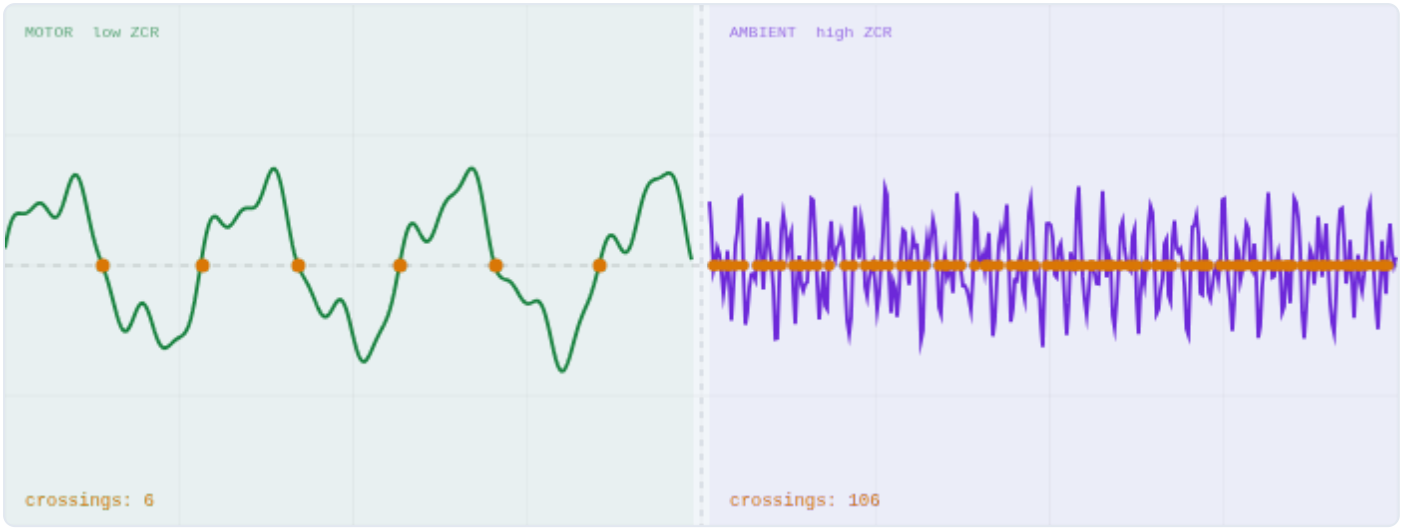
Number of sign changes per unit time — measures signal rhythmicity vs chaos

AMBIENT FILTER

FORMULA

$$ZCR = 1/(2N) \cdot \Sigma \, |sgn(x_n) - sgn(x_{n-1})|$$

sgn = sign function · counts polarity flips per sample



TIME DOMAIN · MOTOR SIGNAL (LEFT, LOW ZCR) VS AMBIENT NOISE (RIGHT, HIGH ZCR) · ZERO-CROSSINGS MARKED

Human speech and crowd noise are **spectrally chaotic** — they cross zero hundreds of times per second because they carry many simultaneous frequencies. An electric motor produces a **rhythmic, periodic signal** with a much lower, predictable ZCR. ZCR is the primary ambient filter: when ZCR is high **and** Kurtosis variance is also high, the brain concludes the source is environmental noise regardless of what RMS reports.

STATE	ZCR LEVEL	COMBINED WITH KURTOSIS Σ	DECISION
● **Healthy Motor**	Low, stable	Low variance	Motor normal — classify
● **Mechanical Fault**	Low + impulse spikes	Low variance	Alert — structural fault
● **Ambient / Crowd**	Very high, variable	**High variance**	Discard — environmental noise
● **Dead Silent**	Near zero	Minimal	Motor off

Predictive Brain — Decision Logic

How the six features combine to produce a final classification label

CLASSIFICATION ENGINE

FEATURE × STATE BEHAVIOUR MATRIX

FEATURE	● HEALTHY	● FAULT	● AMBIENT	● SILENT
01 · RMS	**0.20–0.80**	**> 1.20**	**variable**	< 0.05
02 · Kurtosis	**2.5–3.5 (σ low)**	**> 10 (σ low)**	**3–6 (σ HIGH)**	~3.0
03 · Crest Factor	**1.4–2.5**	**> 6.0**	**high, random**	—
04 · Peak Frequency	**40–75 Hz narrow**	**f* ± sidebands**	**undefined**	none
05 · Spectral Centroid	**low (near f*)**	**elevated ↑**	**very high**	—
06 · Zero Crossing Rate	**low, stable**	**low + spikes**	**VERY HIGH ↑↑**	near zero

COMBINED DECISION RULES

HEALTHY — RMS in [0.2–0.8] + **Kurtosis 2.5–3.5** (low variance) + **ZCR low & stable** + Peak Frequency matches expected voltage mode → Motor running normally.

FAULT — **Kurtosis > 8** with low variance (persistent) + **ZCR rhythmic** + **Crest Factor > 5** → Mechanical fault. If Spectral Centroid also rising → bearing friction imminent.

AMBIENT — **ZCR high** + **Kurtosis variance high** (fluctuating) + Peak Frequency undefined or broadband → Environmental noise. Discard reading, suppress alert.

SILENT — **RMS < 0.05** + all other features near baseline → Motor off or no mechanical activity detected.